NUMERICAL PROBLEMS

Q1. Calculate the line currents in the three-wire Y-Y system of Fig. 1.

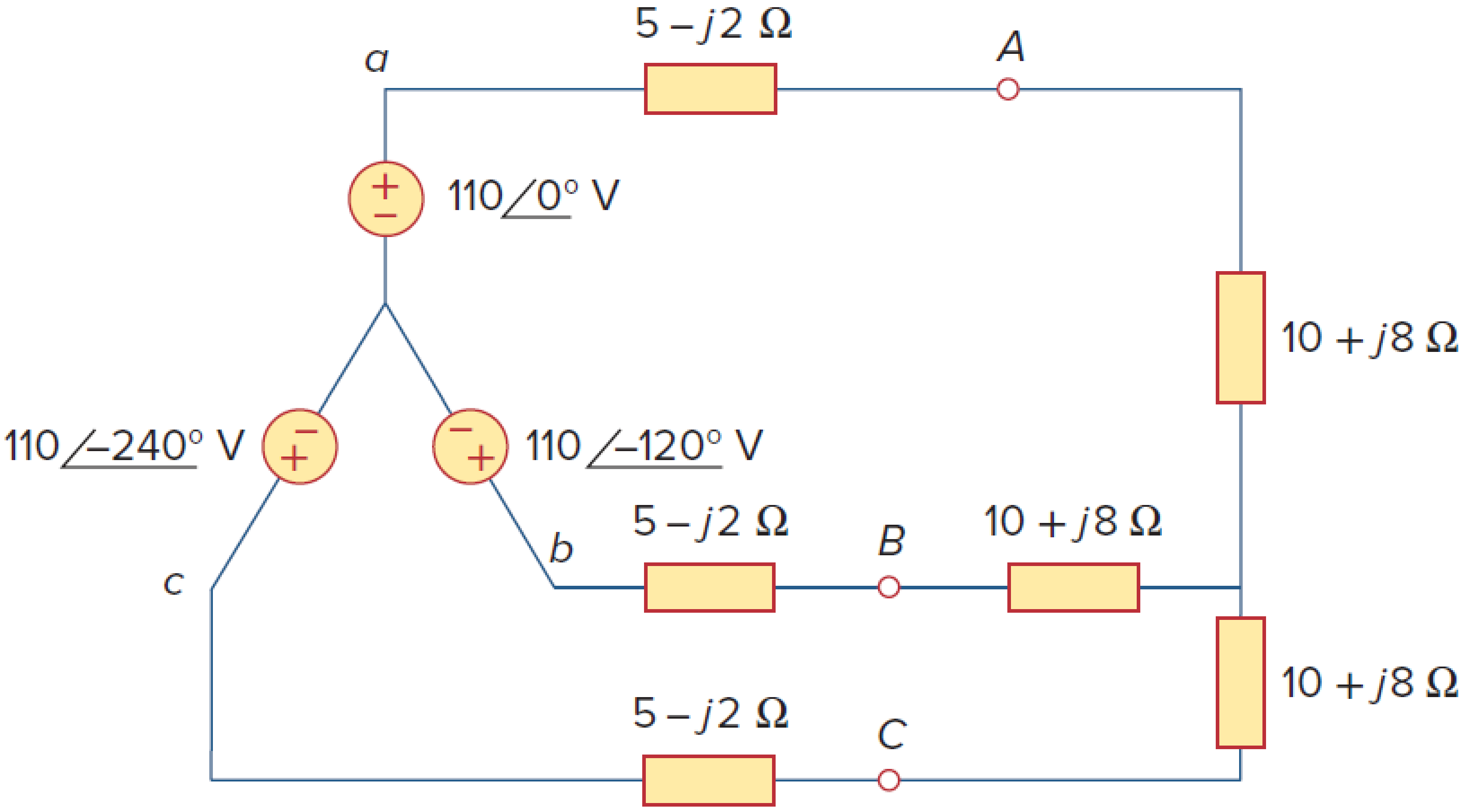


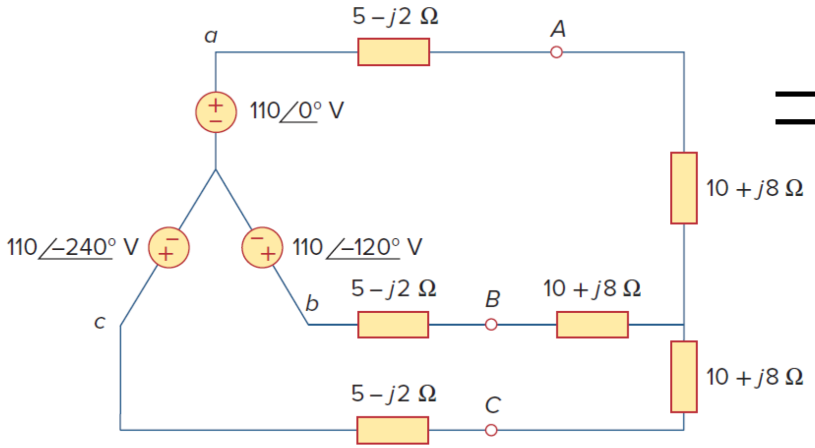
Fig. 1. For Q1.

NUMERICAL PROBLEMS

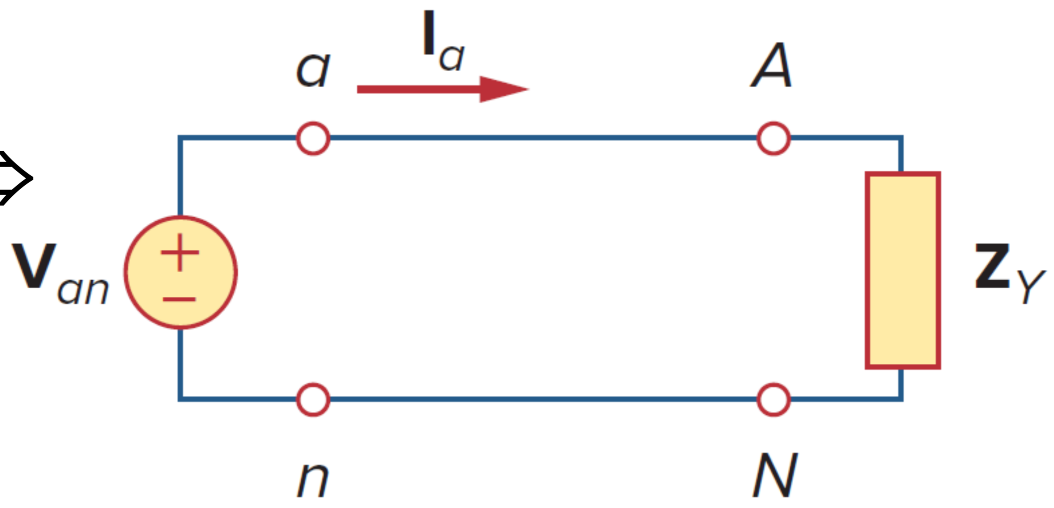
Ans. The three-phase circuit shown in the figure is balanced; we may replace it with its single-phase equivalent circuit.

We obtain I_a from the single-phase analysis as,

$$I_a = \frac{V_{an}}{Z_Y}$$



$\Rightarrow$



A single-phase equivalent circuit of balanced Y-Y connection

where

$$Z_Y = (5 - j2) + (10 + j8) = (15 + j6)$$

$$= 16.155\angle 21.8^0$$

NUMERICAL PROBLEMS

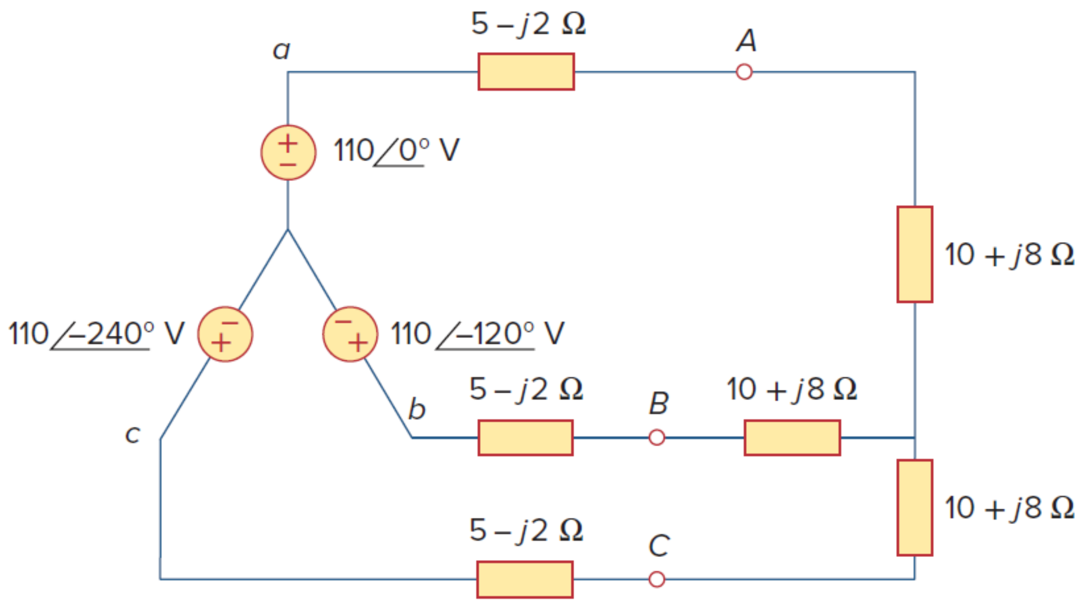
Hence,

$$I_a = \frac{110\angle 0^0}{16.155\angle 21.8^0} = 6.81\angle -21.8^0\,\text{A}$$

In as much as the source voltages in Fig. 1 are in positive sequence, the line currents are also in positive sequence:

$$I_b = I_a\angle -120^0 = 6.81\angle -141.8^0\,\text{A}$$

$$I_c = I_a\angle -240^0 = 6.81\angle 98.2^0\,\text{A}$$



NUMERICAL PROBLEMS

Q1. A balanced *abc*-sequence Y-connected source with $V_{an} = 100\angle 10^0$ V is connected to a Δ-connected balanced load (8 + j4) Ω per phase. Calculate the phase and line currents.

Ans.

The load impedance is:

$$Z_{\Delta} = 8 + j4 = 8.944\angle 26.57^0 \Omega$$

If the phase voltage $V_{an} = 100\angle 10^0$ V, then the line voltage is

$$V_{ab} = V_{an}\sqrt{3}\angle 30^0 = 173.2\angle 40^0 \text{ V} = V_{AB}$$

NUMERICAL PROBLEMS

The phase currents are,

$$I_{AB} = \frac{V_{AB}}{Z_\Delta} = \frac{173.2\angle 40^0}{8.944\angle 26.57^0} = 19.36\angle 13.43^0 \text{ A}$$

$$I_{BC} = I_{AB}\angle -120^0 = 19.36\angle -106.57^0 \text{ A}$$

$$I_{CA} = I_{AB}\angle +120^0 = 19.36\angle 133.43^0 \text{ A}$$

The line currents are,

$$I_a = I_{AB}\sqrt{3}\angle -30^0 = 33.53\angle -16.57^0 \text{ A}$$

$$I_b = I_a\angle -120^0 = 33.53\angle -136.57^0 \text{ A}$$

$$I_c = I_a\angle +120^0 = 33.53\angle 103.43^0 \text{ A}$$

NUMERICAL PROBLEMS

Q1. A balanced Δ-connected load with impedance $20 - j15\ \Omega$ is connected to a Δ-connected, positive-sequence generator having $V_{ab} = 330\angle 0^0$ V. Calculate the phase currents of the load and the line currents.

Ans: The load impedance per phase is $Z_\Delta = 20 - j15 = 25\angle -36.57^0\,\Omega$

Since $V_{AB} = V_{ab}$, the phase currents are,

$$I_{AB} = \frac{V_{AB}}{Z_\Delta} = \frac{330\angle 0^0}{25\angle -36.87^0} = 13.2\angle 36.87^0 \text{ A}$$

$$I_{BC} = I_{AB}\angle -120^0 = 13.2\angle -83.13^0 \text{ A}$$

$$I_{CA} = I_{AB}\angle +120^0 = 13.2\angle 156.87^0 \text{ A}$$

NUMERICAL PROBLEMS

For a delta load, the line current always lags the corresponding phase current by 30° and has a magnitude √3 times that of the phase current. Hence, the line currents are,

$$I_a = I_{AB}\sqrt{3}\angle -30^0 = 22.86\angle 6.87^0 \text{ A}$$

$$I_b = I_a\angle -120^0 = 22.86\angle -113.13^0 \text{ A}$$

$$I_c = I_a\angle +120^0 = 22.86\angle 126.87^0 \text{ A}$$

NUMERICAL PROBLEMS

Q1. A balanced Y-connected load with a phase impedance of 40 + j25 Ω is supplied by a balanced, positive sequence Δ-connected source with a line voltage of 210 V. Calculate the phase currents. Use V_{ab} as a reference.

Ans:

The load impedance is

$$Z_Y = 40 + j25 = 47.17\angle 32^0\,\Omega$$

and the source voltage is

$$V_{ab} = 210\angle 0^0 \text{ V}$$

When the Δ-connected source is transformed to a Y-connected source,

$$V_{an} = \frac{V_{ab}}{\sqrt{3}}\angle -30^0 = 121.2\angle -30^0 \text{ V}$$

NUMERICAL PROBLEMS

The line currents are,

$$I_a = \frac{V_{an}}{Z_Y} = \frac{121.2\angle -30^0}{47.12\angle 32^0} = 2.57\angle -62^0 \text{ A}$$

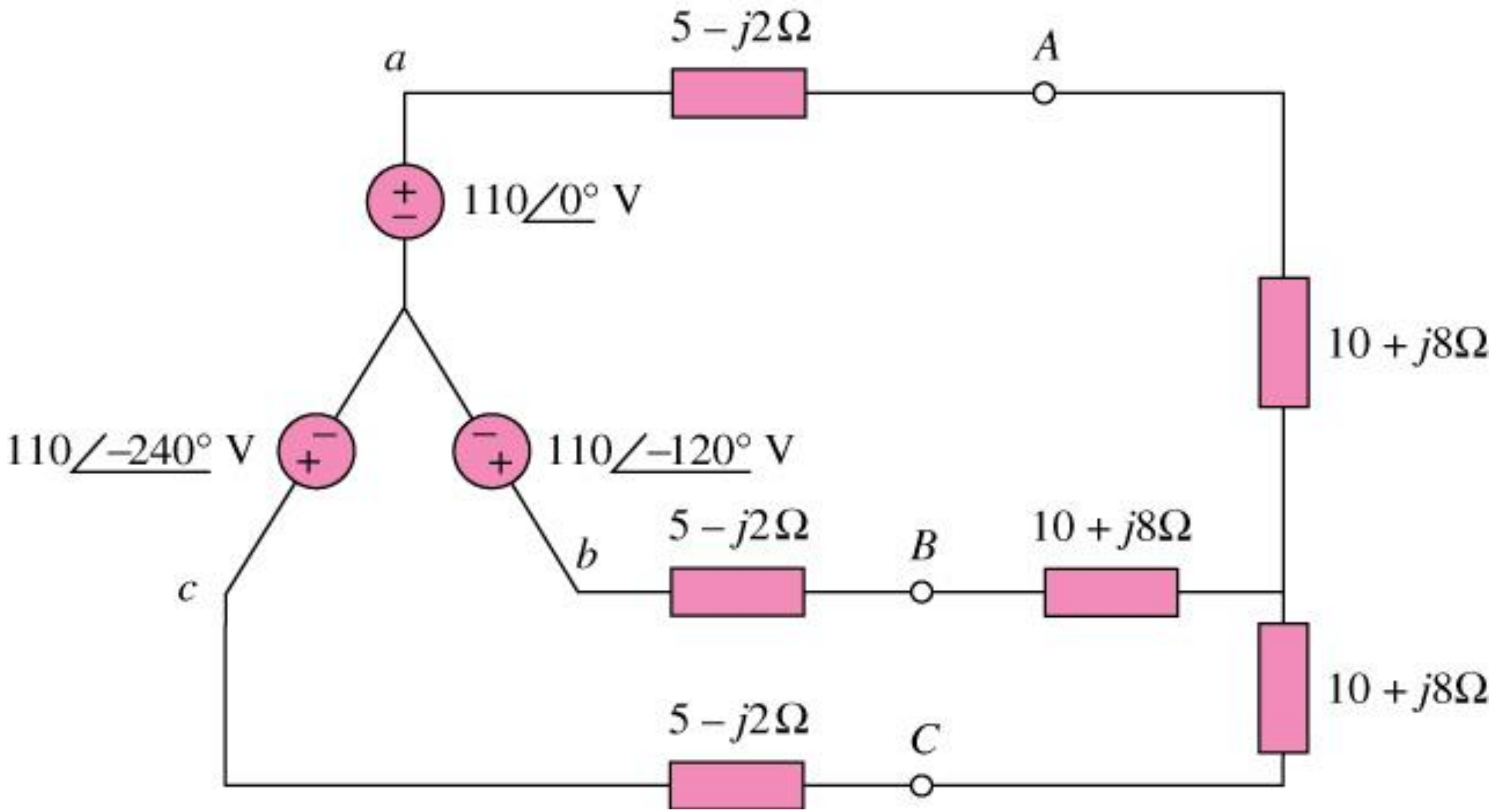
$$I_b = I_a\angle -120^0 = 2.57\angle -178^0 \text{ A}$$

$$I_c = I_a\angle +120^0 = 2.57\angle 58^0 \text{ A}$$

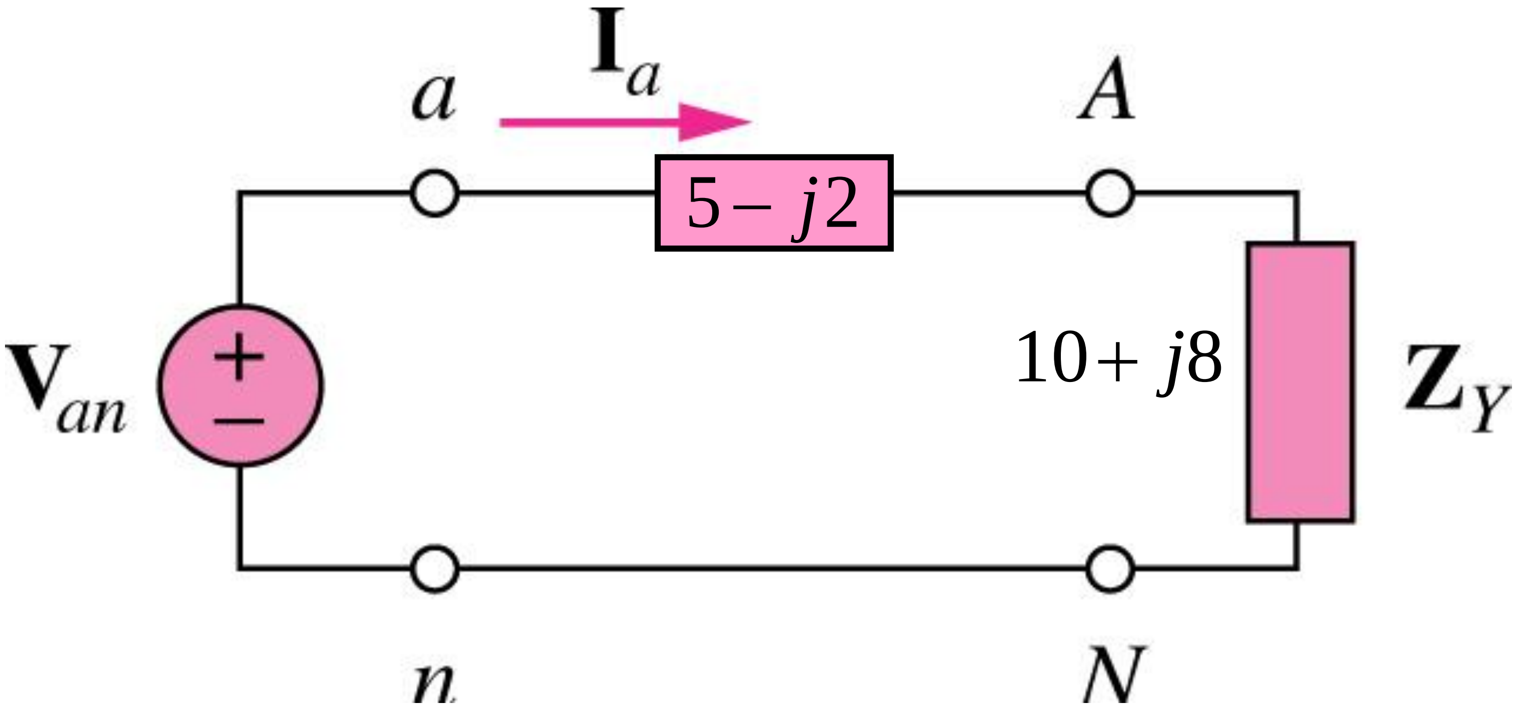
which are the same as the phase currents.

Numerical 1

Calculate the line currents.



Single Phase Equivalent Circuit



Total impedence per phase $Z_Y = 15 + j6 = 16.155\angle 21.8°$

$$I_a = \frac{V_{an}}{Z_Y} = \frac{110\angle 0°}{16.155\angle 21.8°} = 6.81\angle -21.8°$$

$$I_b = I_a\angle -120°$$
$$= 6.81\angle -141.8°\,A$$

$$I_c = I_a\angle -240°$$
$$= 6.81\angle -261.8° = 6.81\angle 98.2°\,A$$

Numerical 2

A balanced delta-connected load having an impedance $20 - j15\ \Omega$ is connected to a delta-connected, positive-sequence generator having $V_{ab} = 330\angle 0^\circ$ V.

Calculate the phase currents of the load and the line currents.

Numerical 2

Solution: $Z_{\Delta} = 20 - j15\ \Omega = 25\angle -36.87°$

$V_{ab} = 330\angle 0°$

Phase Currents:

$$I_{AB} = \frac{V_{AB}}{Z_{\Delta}} = \frac{330\angle 0°}{25\angle -36.87°} = 13.2\angle 36.87° \text{A}$$

$$I_{BC} = I_{AB}\angle -120° = 13.2\angle\text{-}83.13° \text{A}$$

$$I_{CA} = I_{AB}\angle +120° = 13.2\angle 156.87° \text{A}$$

Line Currents:

$$
\begin{aligned}
I_a &= I_{AB}\sqrt{3}\angle -30^\circ \\
&= \left(13.2\angle 36.87^\circ\right)\left(\sqrt{3}\angle -30^\circ\right)A \\
&= 22.86\angle 6.87^\circ
\end{aligned}
$$

$$I_b = I_a\angle -120^\circ = 22.86\angle \text{-}113.13^\circ A$$

$$I_c = I_a\angle +120^\circ = 22.86\angle 126.87^\circ A$$

Numerical 3

A balanced positive sequence Y-connected source with $V_{an} = 100\angle 10°$ V is connected to a Δ-connected balanced load with impedance $(8 + j4)$ Ω per phase.

Calculate the phase and line currents at the load.

Numerical 3

Solution: Balanced Y-source with $V_{an} = 100\angle 10°$ V
Balanced Δ-load with $Z_{\Delta} = 8+j4\ \Omega$

Phase Current $I_{AB} = \dfrac{V_{AB}}{Z_{\Delta}}$

Line Voltage $V_{AB} = \sqrt{3}\ V_{an}\angle 30°$

$V_{AB} = 173.2\angle 40°$ V

$$=> \ I_{AB} = \frac{173.2\angle 40°}{8+j4} = 19.36\angle 13.43°$$

Phase Currents:

$$I_{AB} = 19.36\angle 13.43° \quad A$$

$$I_{BC} = I_{AB}\angle -120° = 19.36\angle -106.57° \quad A$$

$$I_{CA} = I_{AB}\angle +120° = 19.36\angle 133.43° \quad A$$

Line Currents:

$$I_a = \sqrt{3}\ I_{AB}\ \angle -30° = \sqrt{3}\ (19.36)\ \angle(13.43° - 30°)$$

$$I_a = 33.53\ \angle -16.57° \quad A$$

$$I_b = I_a \angle -120° = 33.53\ \angle -136.57° \quad A$$

$$I_c = I_a \angle +120° = 33.53\ \angle 103.43° \quad A$$